1.Problem Statement: Every time is very importnat for cardiac arrest.A medical team can not take early preparation due to lack of data like ECG,SpO2 and blood pressure of the patient,when he is taking to hospital.To overcome this and to increase the chances of saving of patients life,a reliable biotelemetry system is required.This biotelemetry system will transmit the patient's data live from ambulance to the hospitals dashboard so that the doctors can be ready with the necessary equipment in the emergency room.

2.Proposed Model: Our model is divided into 4 parts: Patients and sensors,ambulance,communication and hospital.The proposed model is shown below:

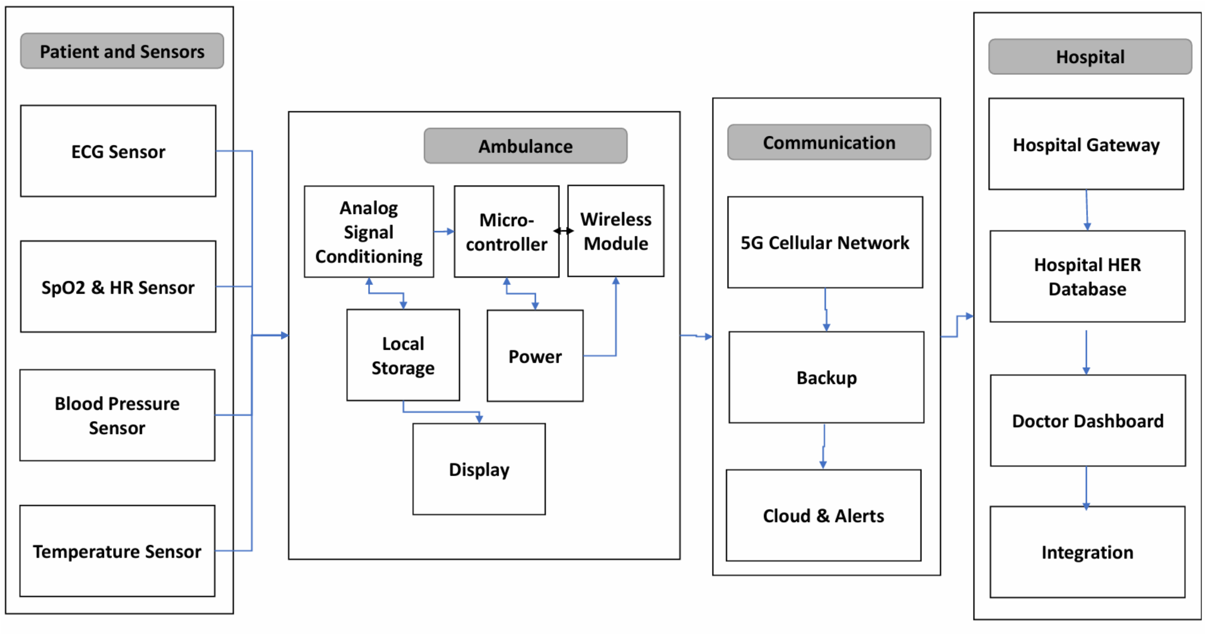

Fig : Overall Model Diagram of Proposed System.

3. Required Components:

a. ECG sensor: AD8232 ECG Module (Operating Voltage 3.3 V, Operating Temperature : -40 to 90 C)

b. SpO_2 & Heart Rate Sensor: MAX30102 (Power Supply Voltage 3.3V-5V, Communication Voltage 1.8-5V)

c. Blood Pressure Sensor: MPX5050 Pressure Sensor + Oscillometric Cuff (Pressure Range: 0–375 mmHg)

d. Temperature Sensor: MLX90614(Operating Voltage 3.6- 5V,Current 1.5mA).

e. Microcontroller Unit: STM32F407VG(Core ARM Cortex-M4, Clock Speed Up to 168 MHz,Flash Memory 512KB,RAM 192KB).

f. Signal Conditioning Circuit: AD620(Operating Voltage 3 – 12V,Offset Voltage 50μV)

g. Wireless Communication Module: SIM8200EA-M2(5G Module,Power adapter 5V,3A).

h. GPS Module: SIM8200EA-M2(Position Accuracy < 2.5 m)

i. Local Data Storage: 32 GB MicroSD Card(Total Flash Capacity:2 GB).

j. DC-DC Converter: LM2596 Buck Converter(input Voltage 4 – 35V,=Output Voltage 3 – 35V)

k. Display: 7-inch TFT LCD(HDMI for displaying, USB for touch)

l. Hospital Gateway Server: Intel i7 Based Medical Server(16 GB RAM, SSD Storag)

m. Doctor Monitoring Station: Dual Monitor Setup,ECG Visualization Software

4. Cost Analysis and Feasibility

The proposed biotelemetry system is designed using commercially available sensors,communication modules,and embedded processing units.The initial implementation cost is higher than conventional patient monitoring systems.

Estimated Hardware Cost

Component	Quan.	Total Cost (BDT)
AD8232 ECG Sensor Module	1	1315
MAX30102 SpO_2 and Heart Rate Sensor	1	545
MPX5050 Pressure Sensor+BP Cuff	1	1525
MLX90614 Temperature Sensor	1	1140
STM32F407 Microcontroller Board	1	2150
AD620 Signal Conditioning Circuit	1	420
SIM8200EA-M2 5G Module	1	40000
SIM7600 4G Backup Module	1	5320
Industrial MicroSD Card(32 GB)	1	950
7 inch TFT Touchscreen Display	1	5650
DC-DC Converter and Power Management Circuit	1	75
Backup Li-ion Battery Pack	1	990
Protective Enclosure & Connectors	1	2000
PCB Fabrication and Assembly	1	3000
	Total	**65080**

5.Economic Feasibility: This system is affordable.It costs about 65000 BDT for each ambulance.This price includes all sensors, the processor,and the 5G module.A normal heart monitor in a hospital costs much more.That can be 550000 to 1650000 BDT.So 65000 BDT is a small cost for an ambulance.The benefits are bigger than the cost.When a patient has a cardiac arrest, every minute matter.If the hospital can prepare before the ambulance arrives,the patient has a better chance to live.Saving one life can save the healthcare system a lot of money. Intensive care and long treatment after a cardiac arrest can cost over 11000000 BDT.If our system helps save even one life per year in a city,it pays for itself many times over.In short,the system is cheap to buy and cheap to run.It gives big value with small price.This makes it a good choice for public ambulance services and private hospitals.

6. **Risk Assessment and Mitigation**:There are some risks in this system.Data can be lost during transmission if the network signal become weak or unavailable.In this case,the hospital may not receive patients information on time.To reduce this problem,data can be stored in a MicroSD card and sent again when the network return.Signal interference may happen because of other electronic devices inside the ambulance.This can make the ECG signal noisy.Shielded wires and digital filters can help reduce this noise.Sensor failure is another risk.An old or damaged sensor may give wrong readings.Using two sensors and checking sensor condition regularly can improve reliability.Power failure can also stop the whole system if the battery runs out.A backup Li-ion battery and power monitoring system can help keep the device running.The communication module may fail because of hardware problems or network issues. Using both 5G and 4G modules provides a backup connection.The microcontroller may stop working due to software errors or memory problems.A watchdog timer and regular health checks can help recover the system.False alarms may occur when the patient moves or when the electrode becomes loose.Signal filtering and checking multiple signals before generating an alarm can reduce this issue.GPS failure can happen when satellite signals are not available. In this situation,A-GPS and mobile tower location can be used as backup methods to track the ambulance location.

7. Ethical and Regulatory Considerations: The biotelemetry system deals with very sensitive patient information so privacy and security is very importance.Patient data should be protected using encryption when sending it through network.Only authorized doctors and hospital staff will be able to access this information.Patient prermission should take before collecting or sharing any medical data whenever it is possible.This system is only for supporting medical decisions and it should not replace the judgment of doctors.Final medical responsibility must stay with qualified healthcare professionals.Since it is used in healthcare,it must follow proper medical device rules and safety standards.It should meet requirements like ISO 13485 for quality management,IEC 60601 for electrical safety,and also follow data protection laws like HIPAA or GDPR.Regular testing,maintenance and security updates are also needed to keep the system safe and reliable in hospitals and emergency services

8. Addressing Complex Engineering Problems:

Attribute	**Requirement**	**Justification**
P1:Depth of Knowledge	Use advanced blocking ideas	Deep Knowledge of biomedical signal processing,embedded systems and 5G telecommunication protocols are used here in a sequence.
P3:Depth of Analysis	First principle analysis of complex problem.	Our architecture was designed analysing real world constraints such as vehicle noise,data security,data loss etc.
P7:Interdependence	Interdependence of equipments	All elements are dependent on each other.This dependency is handeled perfectly to maintain real time performance and accuracy

Conclusion: